A4.1 Evolution and speciation

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 **DEFINE** evolution. [A4.1.1] [1]

2 **STATE** three forms of molecular or anatomical evidence for evolution. [A4.1.2–4] [3]

3 **DEFINE** speciation. [A4.1.6] [1]

4 **NAME** the river that separates bonobos from chimpanzees. [A4.1.7] [1]

Section B: Understanding

13 MARKS

5 **EXPLAIN** why muscles developed through exercise are not an example of evolution. [A4.1.1] [2]

6 **EXPLAIN** how selective breeding provides evidence for evolution. [A4.1.3] [3]

7 **DISTINGUISH** between homologous and analogous structures. [A4.1.4, A4.1.5] [3]

HOMOLOGOUS	ANALOGOUS

8 **EXPLAIN** why similar structures can be misleading when classifying organisms. [A4.1.5] [2]

9 **EXPLAIN** why geographical isolation alone does not guarantee speciation. [A4.1.7] [3]

Section C: Working with data

11 MARKS

All data in this section are invented for practice. The patterns are realistic.

10 [A4.1.2]

The table shows the number of amino acid differences in a section of one protein, compared between pairs of species.

	SPECIES P	SPECIES Q	SPECIES R
HUMAN	1	8	31
SPECIES P	–	9	32
SPECIES Q	9	–	30

a **IDENTIFY** the species most closely related to humans, giving a reason. [2]

b **SUGGEST** which species split from the others first. [1]

c **EXPLAIN** why more differences suggest a longer time since a split. [2]

11 [A4.1.3]

Over 40 generations of selection for oil content in maize, the mean oil content rose as shown.

GENERATION	0	10	20	40
MEAN OIL CONTENT / %	4.5	7.0	9.5	13.5

a **CALCULATE** the mean rise in oil content per generation. [2]

b **EXPLAIN** how the breeders brought about this change. [2]

c **EXPLAIN** why this is not an example of natural selection. [2]

Section D: Longer answer

6 MARKS

12 **EXPLAIN** how one species can split into two, using a named example. [A4.1.6, A4.1.7] [6]